$$\frac{1}{1-u} = 1 + u + u^2 + u^3 + \cdots$$

$$\frac{1}{(1-u)^2} = 1 + 2u + 3u^2 + 4u^3 + \cdots$$

$$= \sum_{r=0}^{\infty} (r+1)u^r$$

When $u = x^6$,

$$(1-x^6)^{-2} = \sum_{r=0}^{\infty} (r+1)x^{6r}$$

(i)

$$(1-x^3)^{-1} = \sum_{s=0}^{\infty} x^{3s}$$

$$6r + 3s = 24 \implies 2r + s = 8 \qquad (r = 0, 1, 2, 3, 4)$$

$$\text{coefficient of } x^{24} = 1 + 2 + 3 + 4 + 5 = 15$$

The coefficient of x^n

$$\begin{cases} 0, & n = 6k + \{1, 2, 4, 5\} \\ \frac{1}{2}(k+1)(k+2), & n = 6k \\ \frac{1}{2}(k+1)(k+2), & n = 6k + 3 \end{cases}$$

(ii)

$$(1-x)^{-1} = \sum_{t=0}^{\infty} x^t$$

For x^{24},

$$6r + 3s + t = 24$$

$$s = 0, 1, \ldots, 8 - 2r, \qquad (r = 0, 1, 2, 3, 4)$$

$$\begin{aligned}\text{coefficient} &= \sum_{r=0}^{4}(r+1)(9-2r)\\ &= 1(9)+2(7)+3(5)+4(3)+5(1)\\ &= 55\end{aligned}$$

For x^{25},

$$6r+3s+t=25$$

$$s=0,1,\ldots,8-2r, \qquad r=0,1,2,3,4$$

$$\text{coefficient}=55$$

For x^{66},

$$6r+3s+t=66$$

$$s=0,1,\ldots,22-2r, \qquad r=0,1,\ldots,11$$

$$\begin{aligned}\text{coefficient} &= \sum_{r=0}^{11}(r+1)(23-2r)\\ &= \sum_{j=1}^{12} j(25-2j)\\ &= 25\sum_{j=1}^{12} j - 2\sum_{j=1}^{12} j^2\\ &= 25(78)-2(650)\\ &= 650\end{aligned}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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